

A MINI PROJECT REPORT ON SHORTEST PATH FINDER USING DIJKSTRA ALGORITHM

Submitted by :

1. Deepak-23AK1A0535
2. Devaraju-23AK1A0536
3. Dinesh - 23AK1A0540

ABSTRACT

This project implements the Dijkstra Algorithm to calculate the shortest path between cities using a Java Swing graphical interface.

INTRODUCTION

The project focuses on shortest path calculation using graph algorithms and visualization techniques.

PROBLEM STATEMENT

Finding shortest routes manually in large networks is difficult and time-consuming.

OBJECTIVES

To design a Java-based application that calculates shortest paths and displays route distances clearly.

SYSTEM DESIGN

The system contains login modules, dashboard interface, graph visualization, and shortest path calculation engine.

TECHNOLOGIES USED

Java, Java Swing, Dijkstra Algorithm, Graph Data Structure, GitHub.

WORKING PROCESS

The user selects source and destination cities. The algorithm calculates and displays the shortest route and distance.

ADVANTAGES

Fast route calculation, user-friendly interface, and efficient graph visualization.

FUTURE ENHANCEMENTS

Real map integration, traffic analysis, and mobile application support.

CONCLUSION

The project successfully demonstrates shortest path calculation using Dijkstra Algorithm.

Technology	Purpose
Java	Programming
Java Swing	GUI
Dijkstra Algorithm	Shortest Path
Graph Data Structure	Graph Processing